

Exploring how the Internal Contingencies of Self-Esteem Contribute to Eating Disorder

Behaviours

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Abstract

Eating disorders remain highly prevalent throughout society, with lifetime prevalence rates exceeding 8% in women. Although previous research has demonstrated the strong reciprocal relationship between eating disorders and self-esteem, self-esteem's multifaceted nature (comprising of both internal and external contingencies) suggests that isolating these specific contingencies will provide a more comprehensive understanding of self-esteem's role in eating disorders. The present study therefore examined how specific internal contingencies of self-esteem relate to eating disorder behaviours. We hypothesised that (H1) higher self-esteem will significantly predict decreases in eating disorder behaviours, (H2) higher difficulties in emotion regulation will significantly predict increases in eating disorder behaviours, and that (H3) higher self-compassion will significantly predict decreases in eating disorder behaviours. Methods: 138 students aged 18 to 25 were surveyed to measure eating disorder behaviours (Eating Disorder Examination Questionnaire version 6.0; EDE-Q), self-esteem (Rosenberg Self-Esteem Scale; RSES), self-compassion (Self-Compassion Scale; SCS), and difficulties in emotion regulation (Difficulties in Emotion Regulation Scale; DERS). Data was analysed using a hierarchical multiple regression, entering the RSES and DERS in step 1, and self-compassion in step 2. Results: Of the three hypothesised predictors, only self-compassion significantly predicted eating disorder behaviours ($B = -0.72$; $\beta = -.40$), supporting H3 but not H1 or H2. While the RSES and DERS accounted for 12.8% of the variance in EDE-Q scores, this increased by another 13.9% (to a total of 26.7%) when the SCS was added, displaying its independent effect on eating disorder behaviours. Conclusion: Although significantly correlated as individual predictors, self-esteem and difficulties in emotion regulation lost their unique predictive power when self-compassion was included, suggesting a potential mediation or suppression effect. This demonstrates that, among the internal contingencies of self-esteem examined, only self-compassion uniquely predicted

prevalence and severity of eating disorder behaviours, highlighting its role as a proximal affect-regulation mechanism that soothes negative emotions and reduces self-criticisms. The present study therefore demonstrates the potential for self-compassion focused clinical interventions when treating and preventing eating disorder behaviours.

Keywords: eating disorders, self-esteem, internal contingencies, self-compassion, difficulties in emotion regulation

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Eating disorders (ED) refer to behavioural conditions characterised by severe and persistent disturbances in eating behaviours and associated distressing thoughts and emotions, significantly impairing physical and psychosocial functioning (*American Psychiatric Association, 2013*). EDs have a prevalence of 8.4% for women's and 2.2% for men's mean lifetime risk of development, therefore constituting a major public health concern (Galmiche et al., 2019). There are several types of EDs, including Anorexia Nervosa (AN), Bulimia Nervosa (BN), and Binge Eating Disorder (BED), which are characterised by disordered eating patterns, distorted body image, and severe psychological distress. The development and maintenance of these disorders is influenced through a complex interplay of biological, psychological, and sociocultural factors, going beyond just body image concerns.

Self-esteem, referring to the level in which one's self-concept is perceived positively or negatively (*American Psychological Association, 2023*), plays a crucial role in mental health, and more specifically in ED behaviours, with prior research showing a significant link between low self-esteem and an increased risk of developing an ED, as seen through studies utilising the Rosenberg Self-Esteem Scale (RSES; Rosenberg, 1979) that highlight a statistically significant relationship between low self-esteem and elevated scores on the EAT-26 (Eating Attitudes Test 26; Mallaram et al., 2023). Furthermore, a reciprocal relationship has been demonstrated between the two in both short and long-term studies (Krauss et al., 2023; Beckers et al., 2024), highlighting the dynamic relationship between self-esteem and EDs over time.

Despite clear links between ED behaviours and self-esteem, existing literature often treats self-esteem as a unidimensional measure. However, due to its multidimensional nature (Shavelson, Hubner, & Stanton, 1976; Rentzsch, Wenzler, & Schutz, 2016) this approach fails to provide a comprehensive understanding of the different factors that can influence ED behaviours. Factors that influence self-esteem, such as perfectionism (Stackpole et al., 2023),

self-compassion (Turk & Waller, 2020), and emotional regulation (Polivy & Herman, 1993) have all been identified as strong predictors of ED behaviours, highlighting the limitations of using generalised measures like the RSES.

Acknowledging this complex nature, it becomes important to distinguish between the internal and external factors that contribute an individual's self-esteem. While external factors, (e.g., appearance and social influences), have received considerable attention through past research (e.g., Frieiro et al., 2022), the present study specifically explores internal factors and their unique influence on ED behaviours, allowing for the development of a more comprehensive understanding of the cognitive factors that shape an individual's self-esteem, and in-turn the role these play in ED behaviour.

When differentiating between internal and external factors that contribute to an individual's self-esteem, the present study utilises the Contingencies of Self-Worth framework (Crocker & Wolfe, 2003) which establishes that the aspects of self-esteem fall on a continuum, ranging from internal (deriving from personal standards and beliefs) to external (deriving from the validation of others). Acknowledging this, the present study proceeds by referring to these internal factors of self-esteem as internal contingencies.

The first key internal contingency of self-esteem selected within the present study was emotion regulation (or more specifically, difficulties in emotion regulation), referring to the processes in which individuals influence which emotions they experience, when they experience them, and how they experience and express them (Gross, 1998). Within Gross's process model, individuals use varying strategies (such as cognitive reframing of an event) to regulate their emotions, and as these are said to depend on an individual's own abilities (self-control and virtue, rather than external validation) emotional regulation is therefore seen to be an internal contingency.

The second key internal contingency of self-esteem selected within the present study is self-compassion, which refers to the dynamic balance between compassionate and uncompassionate ways in which individuals emotionally respond to pain and failure, understand their own difficulties, and pay attention to their suffering (Neff, 2015). As self-compassion is seen to primarily derive from individuals' standards of themselves it can be seen to fall under the internal contingency bracket, influencing self-esteem without depending on appearance, achievement, or validation from others.

Thus, the present study adopts a cross-sectional design to examine whether two internal contingencies, self-compassion and difficulties in emotion regulation, uniquely predict ED behaviours above that of self-esteem, within in a non-clinical student sample. By focusing on the internal contingencies of self-esteem, the present study directly addresses gaps left by earlier work that either overlooked internal contingencies, or treated self-esteem as a unidimensional measure, allowing for a deeper understanding of the mechanisms that influence ED behaviour to be gained. Furthermore, these internal contingencies have been shown to be directly modifiable through treatment approaches such as Compassion-focused therapy (Gilbert, 2009; Vidal & Soldevilla, 2022) and Dialectical Behaviour Therapy (Linehan, 1993; Azevedo et al. 2024), further justifying the focus placed on them within the present study. Therefore, the present study's findings will potentially improve awareness surrounding EDs, as well as inform clinical interventions, by highlighting the areas that more personalised treatment methods should focus on.

Based on existing research such as the Core Beliefs Model of Disorder Eating (Hatoum, Burton, & Abbott, 2024), which states that "core low self-esteem" correlates with increases in binge-eating and dietary-restraint, as well as previous findings displaying a negative correlation between the SES (self-esteem scale) and EAT-26 (Pelc et al., 2023), the present study first hypothesises (H1): Higher self-esteem (as measured by the RSES) will

significantly predict decreases in ED behaviours (as measured by the EDE-Q 6.0; Fairburn & Beglin, 2008).

Secondly, based on the Affect Regulation Model of Binge Eating (Heatherton & Baumeister, 1991) as well as previous research displaying a significant positive correlation between difficulties in emotion regulation and more severe eating-disorder pathologies ($r = 0.36$; Leppanen et al., 2022), the present study hypothesises (H2): Higher difficulty in emotion regulation (as measured by the Difficulties in Emotion Regulation Scale; DERS; Gratz & Roemer, 2004) will significantly predict increases in ED behaviours.

Lastly, based on previous research into self-compassion and EDs that reveals a significant negative correlation between self-compassion and ED pathology ($r = -0.34$; Turk & Waller, 2020), the present study hypothesises (H3): Higher self-compassion (as measured by the Self-Compassion Scale; SCS; Neff, 2003) will significantly predict decreases in ED behaviours.

Methods

Participants

Participants were all University students, recruited through the “Participating in Research” University of Sussex Padlet and study swap email invitations, who voluntarily took part in the study and were entered into a random draw for a £25 e-voucher. All participants were provided with a participant information statement containing information regarding the aims of the study, requiring them to provide acknowledgment of their understanding and consent to participate in the present study.

Of the 183 participants that responded to the survey, 138 participants completed all measures of interest to a sufficient level (EDE-Q, RSES, DERS, or SCS), and were subsequently included in the data analysis (Female = 103, Male = 35). Participants ranged from 18 to 25 years of age ($M = 21$) and were recruited between the 27th of January and the 28th of March 2025.

Exclusion Criteria

Responses displaying fewer than 80% completion rates for any core measure were removed from the final dataset, as this threshold is commonly recommended to ensure high construct validity and minimise non-response bias (Bennett, 2001; Schafer and Graham, 2002). Responses falling below this cut-off point would provide too little information to allow for stable scores or trustworthy correlations to be observed; therefore 45 responses were excluded prior to analysis. No additional exclusion criteria were applied to the dataset.

Materials

Online survey. To assess whether self-esteem, self-compassion, and difficulties in emotion regulation predicted ED behaviours a series of self-report questionnaires were

administered via Qualtrics, an online survey platform (*Qualtrics software, Version 01/2025 of Qualtrics. Copyright © 2025 Qualtrics. Qualtrics and all other Qualtrics product or service names are registered trademarks or trademarks of Qualtrics, Provo, UT, USA.* <https://www.qualtrics.com>).

In the present study four measures, operationalised using self-report scales, were administered to examine the areas of interest: ED behaviours (*EDE-Q 6.0*), self-esteem (RSES), self-compassion (SCS), and difficulties in emotion regulation (DERS-36). Each scale was selected based on their strong psychometric properties and relevance to the present study's aims.

The Eating Disorder Examination Questionnaire version 6.0 (EDE-Q 6.0). The EDE-Q is a self-report questionnaire composed of 28 items, used to assess the frequency of key behaviours such as binge-eating, restriction, and self-induced vomiting, as well as the severity of other ED symptoms. It utilises a 7-point Likert scale (ranging from “0 = no days” to “6 = every day”) to determine how frequently these behaviours occur over the past 28 days, with higher scores indicating more severe eating difficulties. A sample item is: “Have you had a definite fear that you might gain weight?” (*item-10*). It is widely used in both clinical and non-clinical samples, demonstrating strong reliability, and good convergent and concurrent validity with more recent ED measures such as the ED-15 (Eating Disorder-15; Grilo, Masheb & Wilson, 2015).

While the EDE-Q is traditionally scored as a mean of the four sub-scales and a global mean, the present study analysed the raw totals of the 28 items (0-168) for several reasons. Recent psychometric work has displayed the total scale as essentially unidimensional, with the present study displaying higher reliability for total scores than subscales (present study $\alpha = .94$ vs. $\alpha = .78 - .93$; Laskowski et al., 2023). Additionally, exploratory factor analysis showed acceptable fit and strong factor score adequacy, and a bifactor analysis produced

strong general-factor loadings. Furthermore, each subscale strongly correlated with total scores ($r = .84-.93$), indicating that summed total scores serve as a more robust measure, with no substantial change to interpretation (Field, 2013). Overall, the EDE-Q 6.0's strong psychometrics, excellent reliability, and unidimensionality support its use as a self-report measure of ED behaviours in the present study.

Rosenberg Self-Esteem Scale (RSES). The RSES is a self-report questionnaire composed of 10 items (with total scores ranging from 10-40, items scored 1-4), used to measure self-esteem. It is seen to be the standard measure of self-esteem in psychological research, utilising a 4-point response format ranging from “strongly agree” to “strongly disagree”, with higher scores indicating higher self-esteem. A sample item is “I feel that I have a number of good qualities” (item-3). It also exhibits solid psychometrics, with strong to excellent reliability ($\alpha = .77$, present study $\alpha = .80$; test-retest $r = .82$), a unidimensional factor structure, and moderate convergent validity with academic self-concept ($r = .44$; Rosenberg, 1987; Fleming & Courtney, 1984; Hensley, 1977; Reynolds Ramirez, Magrina, & Allen, 1980). The RSES was selected as its well-established psychometric strengths confirm it as a reliable and valid self-report instrument for capturing trait self-esteem, directly aligning with the present study's first predictor, self-esteem.

Self-Compassion Scale (SCS). The SCS is a self-report questionnaire composed of 26 items (13 of which are negatively worded and scored; 1, 2, 4, 6, 8, 11, 13, 16, 18, 20, 21, 24, 25), used to measure levels of self-compassion, comprising of a further six subscales; measurements of self-kindness, self-judgement, common humanity, isolation, mindfulness, and over-identification. It utilises a 5-point Likert scale ranging from “almost never” to “almost always”, with higher scores indicating greater self-compassion. A sample item is “I'm tolerant of my own flaws and inadequacies” (item-23). It is widely used self-report

measure with excellent reliability ($\alpha = .92$, present study $\alpha = .89$; test-retest $r = .93$) and acceptable subscale reliability ($\alpha = .75-.81$; Neff, 2003).

Although sometimes scored as a mean of every item, the SCS was originally developed as a single unidimensional construct and scored using a sum of the 26 items (Neff, 2003). A bifactor analysis supports this, displaying an excellent fit (CFI = .99, TLI = .097, RMSEA = .005; Neff, 2007) justifying the use of a single summed total score (26-130) in the present study. The SCS was selected because its strong internal consistency and convergent validity with similar measures establishes it as a robust measure of self-kindness, common humanity, and mindfulness, directly aligning with the second predictor, self-compassion.

Difficulties in Emotion Regulation Scale (DERS-36). The DERS is a self-report questionnaire composed of 36 items (scored between 36-180), used to measure emotion regulation abilities, comprising of a further six subscales: nonacceptance of emotion responses, difficulties engaging in goal-directed behaviour, impulse control difficulties, lack of emotional awareness, limited access to emotion regulation strategies, and lack of emotional clarity. It utilises a 5-point Likert scale ranging from “almost never” to “almost always”, with higher scores indicating greater difficulties in emotion regulation. A sample item is “I pay attention to how I feel” (*item-2*). It is a widely used measure, demonstrating excellent reliability for the full scale ($\alpha = .93 - .95$, present study $\alpha = .89$; test-retest $r = .88$), good to excellent internal consistency for the subscales ($\alpha = .80 - .92$, test-retest $r = .57-.89$), an acceptable factor structure (CFI = .91), and strong convergent validity with the Negative Mood Regulation Scale ($r = -.69$) (Gratz & Roemer, 2004; Fowler et al., 2014). The present study used the DERS-36 due to its strong reliability and in-depth measurement, making it a psychometrically robust and comprehensive instrument for assessing difficulties in emotional regulation which aligns with the present study’s predictor, difficulties in emotion regulation.

Analysis software. Data was analysed using RStudio (*RStudio Team (2020). RStudio version 4.5.0: Integrated Development for R. RStudio, PBC, Boston, MA, URL <http://www.rstudio.com/>*).

Design

The present study utilises a cross-sectional correlational design to examine the relationship between internal psychological factors of self-esteem and ED behaviours, gathering quantitative data, allowing for an in-depth analysis of the predictive relationships in a non-clinical sample.

The predictor variables for the present study are self-esteem (measured using the RSES), difficulties in emotion regulation (measured using the DERS), self-compassion (measured using SCS). The outcome variable is ED behaviours, measured using the EDE-Q.

Procedure

Participants were provided with an information and consent form that specified the study's aims, procedures, potential risks, and benefits, in which they had to select all check boxed confirming they understood, before proceeding to take part in the study.

Participants then went on to complete the online survey, taking approximately 20 minutes and involving 4 different questionnaires condensed into one fixed block. If participants found any questions particularly distressing, they had the ability to skip it and go on to the next (with the score for said question being recorded as '0').

After completing the survey, participants received a debrief sheet that explained the study's purpose, let them know what would happen next, and offered consolation for any potential distress they faced, going on to provide the links and contact information to ED and

mental health support services. Contact information for the researchers was also included in the debrief sheet if participants wished to learn more about the study they participated in.

After reading the debrief sheet, participants were then redirected to a separate Qualtrics survey, in which they could input their contact information to then be entered into the £25 gift card raffle, if they so desired.

Ethical Issues

All procedures carried out in the present study were approved by The University of Sussex Science & Technology or Social Sciences & Arts Research Ethics Committee (project code ER/MG699/1), and in accordance with the British Psychological Society (BPS Code of Human Research Ethics, 2021).

Participants received a full information form, explaining study aims, potential distress due to the sensitive nature of ED items, and all their rights. Consent was provided via a 10-item checklist. To avoid distress, participants could skip any items they chose and were able to withdraw at any point without specifying the reason. All data was anonymised using randomly generated ID codes through Qualtrics, stored on password-protected systems only accessible by the researchers, and will be deleted after five years in accordance with University of Sussex regulations. On completion, a debrief form provided reassurance for, and normalised, any distress experienced, with contact information for support services specified to aid any participants particularly affected.

Data Analysis

We generated descriptive statistics and Cronbach's α for each scale, then inspected normality, homoscedasticity, multicollinearity, independence of errors, and influential points. Zero-order Pearson correlations were followed by a hierarchical multiple regression, in which

RSES and DERS were entered in Step 1 predicting the EDE-Q total, and the SCS added in Step 2. The change in variance was then calculated.

Results

Of the 183 participants (Female = 103, Male = 80) who began the survey by clicking the link, 45 (24.6%) were excluded for not meeting study criteria, as they failed to answer more than 20% of the items on one or more questionnaires. The final sample therefore comprised of 138 participants ($N = 103$ (74.6%) female, $N = 35$ (25.4%) male).

Descriptive Statistics

Mean scores showed low-to-moderate eating-disorder symptoms (EDE-Q), moderate self-esteem (RSES), moderately high self-compassion (SCS), and moderate emotion-regulation difficulties (DERS) in the sample (*see Table 1*). Internal consistencies were good to excellent for the scales (*see Table 6*). Pearson's correlations indicated that none of the study variables correlated significantly with age (largest $r = -.10$; *see Table 4*) and no gender differences emerged (*see Table 5*).

Exploratory Analysis

Exploratory factor analysis of EDE-Q items suggested a unidimensional structure (RMSR = .07; TLI = .88; RMSEA = .08, 90% CI [.05, .12]) and a strong factor structure (CFI = .91; $R^2 = .83$). Parallel analysis favoured a single factor (eigenvalue = 2.91) that explained 32% of the variance. Factor-score determinacy was high ($r = .91$), supporting use of the total EDE-Q score (*see Appendices*).

Within the EDE-Q sub-scales, internal consistency was acceptable for Eating-Concern ($\alpha = .79$), Shape-Concern ($\alpha = .82$), and Weight-Concern ($\alpha = .81$), but poor for Restraint ($\alpha = .59$; *see Table 6*).

Exploratory analysis of the RSES in comparison to the other predictors revealed a positive correlation with the SCS ($r = .26$) and negative correlation with the DERS ($r = -.63$).

Assumption Checks

Inspection of residual-versus-fitted, Q-Q, and Cook's-distance plots (< 0.14) confirmed that regression assumptions were met, and removing the cases with highest Cook's distance did not affect significance or coefficient size (14, 26, 31). Variance inflation factors were 1.14–1.77, indicating acceptable collinearity among predictors. A Breusch-Pagan test indicated no significant heteroscedasticity ($p = .080$), and leverage diagnostics were satisfactory, suggesting homoscedasticity. The Shapiro-Wilk test suggested that EDE-Q 6.0 scores deviated from normality ($W = 0.95, p < .001$), however, inspection of the Q-Q plot showed only mild tail skew, and bootstrapped regression yielded the same pattern of significance, so untransformed scores were retained. Collinearity was acceptable for self-esteem ($VIF = 1.68$), self-compassion ($VIF = 1.14$) and difficulties in emotion regulation ($VIF = 1.77$).

Regression Analysis

Zero-order correlations showed that self-compassion, self-esteem and difficulties in emotion regulation each related significantly to ED severity (SCS $r = -.47$, RSES $r = -.31$, DERS = .33; $p < .001$).

A hierarchical multiple regression analysis then tested whether self-compassion predicted ED behaviours over self-esteem and difficulties in emotion regulation. Self-compassion emerged as the only unique predictor of ED behaviours after accounting for self-esteem and difficulties in emotion regulation, which uniquely explaining 13.9% of outcome variance (see Table 2 and Table 3).

A one-point increase associated with a $-.72$ -point decrease in ED behaviours ($B = -0.722$ displaying a moderate unique effect on ED behaviours, while the unique effects of RSES ($B = -1.00$) and DERS ($B = 0.212$) were not statistically reliable (see Table 2).

Table 1

Descriptive Statistics and 95% CIs for Study Variables (N = 138)

Variable	$M \pm SD$	95% CI for M
Eating Disorder Examination Questionnaire (EDE-Q 6.0)	62.64 $\pm$ 31.55	57.33, 67.95
Self-Compassion Scale (SCS)	68.78 $\pm$ 17.36	65.85, 71.70
Rosenberg Self-Esteem Scale (RSES)	15.96 $\pm$ 4.37	15.22, 16.69
Difficulties in Emotion Regulation Scale (DERS)	100.78 $\pm$ 16.26	98.04, 103.52

Table 2

Hierarchical Regression (Step 1 vs. Step 2)

Variable	B	β	$SE\ B$	t	p	95% CI
Step 1						
(Intercept)	36.929	–	29.510	1.251	.213	-21.433, 95.291
RSES	-1.194	-0.165	.751	-1.591	.114	-2.679, .290
DERS	0.444	.229	.202	2.203	.029	.046, .843
Step 2						
(Intercept)	106.910	–	30.506	3.505	< .001	46.575, 167.244
RSES	-1.002	-0.139	.692	-1.449	.150	-2.371, .366
DERS	0.212	.109	.191	1.111	.269	-0.166, .590
SCS	-0.722	-0.397	.143	-5.038	< .001	-1.006, -0.439

Table 3*Hierarchical Regression Summary (Step 1 vs. Step 2)*

Step	Predictors	R^2	ΔR^2
Step 1	RSES + DERS	.128	—
Step 2	RSES + DERS + SCS	.267	.139

Table 4*Descriptive Statistics and Age Correlations*

Variable	M	SD	95% CI for M	r Age	p
Age	20.78	2.08	20.43, 21.13	—	—
EDE-Q total	62.64	31.55	57.33, 67.95	-.10	.243
Self-Compassion Scale	68.78	17.36	65.85, 71.70	-.05	.591
Rosenberg Self-Esteem	15.96	4.37	15.22, 16.69	.08	.337
DERS total	100.78	16.26	98.04, 103.52	-.06	.494

Table 5*Male vs. Female Descriptive Statistics*

Variable	Female $M \pm SD$	Male $M \pm SD$	t	95% CI	p
Age	20.67 $\pm$ 1.92	21.11 $\pm$ 2.51	.79	-0.29, 1.04	.433
EDE-Q total	64.94 $\pm$ 32.25	55.86 $\pm$ 28.78	-1.56	-20.69, 2.52	.123
Self-Compassion Scale	67.36 $\pm$ 17.62	72.94 $\pm$ 16.10	1.73	-0.86, 12.03	.088
Rosenberg Self-Esteem	15.84 $\pm$ 4.26	16.29 $\pm$ 4.73	.49	-1.37, 2.25	.627
DERS total	101.54 $\pm$ 14.77	98.54 $\pm$ 20.11	-0.81	-10.44, 4.44	.421

Table 6*Internal Consistency for Scales and Subscales*

Scale / Sub-scale	α
EDE-Q total (28 items)	.94
Restraint	.59
Eating Concern	.79
Shape Concern	.82
Weight Concern	.81
RSES	.8
SCS	.89
DERS	.89

Discussion

The aim of the present study was to examine how the internal contingencies of self-esteem (these being self-compassion and emotion regulation) contribute to ED behaviour, with the results displaying that only self-compassion emerged as strong predictor of ED behaviours, whereas difficulties in emotion regulation and self-esteem showed no independent effect once the exclusion criteria had been applied, supporting H3, but not H1 or H2.

These findings align with the Self-Compassion Model (*Neff, 2003*), further expanding its focus from general well-being, by applying it to ED behaviours, broadening the model's range of application and providing clear rationale for targeting self-compassion in clinical interventions for EDs. Furthermore, the present study's hierarchical regression analysis shows how high self-compassion acts as a buffering mechanism, reducing the likelihood of predisposing risk factors (low self-esteem and difficulties in emotion regulation) manifesting as ED behaviours.

Furthermore, these results refine the Contingencies of Self-Worth framework (Crocker & Wolfe, 2003) by demonstrating, among the internal contingencies tested, that self-compassion emerges as the primary mechanism through which self-esteem's protective effect against eating-disorder behaviours acts. Although zero-order correlations showed that self-esteem and difficulties in emotion regulation correlated with ED behaviours when in isolation, their effects were lost once self-compassion was accounted for, implying self-esteems influences eating behaviours via self-compassion.

Self-esteem

Contrary to H1, self-esteem did not display a significant correlation with ED behaviours in the regression model. This finding refines the direct links reported within

previous literature and research that state self-esteem directly correlates with ED behaviours (Button, Sonuga-Barke, Davies & Thompson 1996), alternatively suggesting that the effect of self-esteem on ED behaviour may operate indirectly via self-compassion and difficulties in emotion regulation.

An exploratory analysis of the RSES offered further insight into this interpretation, displaying that self-esteem positively correlated with self-compassion and negatively correlated with difficulties in emotion regulation, replicating previous research findings that show higher self-compassion promotes increased levels of stable self-esteem (Neff & Vonk, 2009), as well as that increased difficulties in emotion regulation drive negative self-appraisals leading to lower self-esteem (Kring & Werner, 2004; Crocker & Wolfe, 2001). These patterns therefore imply that self-esteem may have an influence on ED behaviour even though a direct correlation was not observed in the results, potentially acting as an indirect influencer through self-compassion

Difficulties in emotion regulation

Contrary to H2, the DERS emerged as a non-significant predictor of ED behaviour. This finding contrasts previous research and theory such as the Affect Regulation Model (Heatherton & Baumeister, 1991), which suggests that individuals may use ED behaviours such as bingeing and restriction to regulate emotions, as well as previous research into emotion regulation and ED behaviour that found difficulties in emotion regulation consistently correlated with eating-disorder pathology (Leppanen et al., 2022).

One interpretation that explains this non-significant result can be found through Gross's (1998) Process Model of Emotion Regulation, which states there are five strategies through which emotion can be regulated; selection of the situation, modification of the situation, deployment of attention, change of cognitions, and modulation of response. As total

DERS scores reflect an amalgamation of these strategies, it is possible that the specific regulatory strategies more closely associated with ED behaviours are concealed, further supported through previous meta-analysis research showing that rumination (attentional deployment) had a moderate correlation with EDs, whereas reappraisal (cognitive change) had a weak correlation with EDs (Aldao, Nolen-Hoeksema, Schweizer, 2010; Nolen-Hoeksema et al., 2007). Therefore, by combining all emotion regulation strategies into a single total, the DERS may fail to isolate the strategies with strong correlations, as any strong effect would be obscured by potentially unrelated measures, explaining the non-significant finding.

Alternatively, another interpretation of the non-significant result is that ED behaviours (specifically BED and BN) may act as emotion regulation strategies in their own way (Macht, Haupt, & Ellgring, 2005). Furthermore, Escape Theory poses that binge episodes act to reduce negative distress by narrowing attention from higher level thinking towards immediate sensory experiences (Baumeister, 1990), providing escape from aversive self-awareness (Heatherton & Baumeister, 1991), which cannot be directly captured through the DERS. Therefore, as the DERS measures internal, cognitive-emotional processes involved in emotion regulation, rather than external, behavioural mechanisms, individuals who rely on eating-disorder behaviours to regulate emotions may display relatively normal scores, accounting for the non-significant result.

Self-compassion

Consistent with H3, the SCS emerged as a significant predictor of ED behaviour. This finding aligns with previous meta-analysis research (Turk & Waller, 2020), as well as the Transactional Model of Stress and Coping (Lazarus & Folkman's, 1984), which states that there are three stages individuals engage in when facing a stressor, these being; primary

appraisals (interpretation of the stressor), secondary appraisals (perceived coping ability), and coping effort (overcoming through problem or emotion focused techniques). Within this model, self-compassion is seen to operate through two key pathways.

Firstly, it functions as an emotion-focused coping mechanism, in which individuals engage in soothing behaviours to manage emotional distress, with higher levels of self-compassion correlating with decreases in negative ruminations and increases in emotion regulation ability (Neff & Germer, 2013). Similarly, Arch et al., (2014) reported that short-term increases in self-compassion led to a 23% reduction in cortisol reactivity to stressful events, displaying self-compassion's direct effect on biological stress responses.

Secondly, self-compassion also reshapes secondary appraisals by directly boosting perceived coping resources and abilities, with previous research demonstrating higher levels of self-compassion correlated with greater perceived control over stressors, acting as a mediator between stress and anxiety (Chishima et al., 2018). Moreover, high self-compassion correlated with lower threat ratings of ambiguous stimuli, indicating it also indirectly shapes primary appraisals by reducing self-criticism, in-turn making stressors appear less threatening. A meta-analysis of 136 studies displayed that higher self-compassion negatively correlated with perceived stress (Ewet, Vater, & Schröder-Abé, 2021), with higher levels moderately increasing the likelihood of individuals engaging in adaptive responses (positive coping strategies) and significantly decreasing the likelihood of engagement in maladaptive coping responses (negative coping strategies), supporting self-compassions role in secondary appraisals. Together, these findings support self-compassion acting as both an emotion-focused coping strategy and an influencer of secondary appraisals, while also indirectly shaping primary appraisals, explaining the SCS as a significant predictor within the results through its influence on how threats and coping resources are evaluated.

Internal Contingencies and Self-Esteem

Although the present study did not initially set out to directly test self-esteem in relation to our chosen internal contingencies, an exploratory analysis showed that both self-compassion and difficulties in emotion regulation strongly correlated with self-esteem.

When looking at self-compassion there was a significant positive correlation with self-esteem, meaning participants who had higher self-compassion were more likely to have higher self-esteem; consistent with previous research such as Neff's (2015) self-compassion model that states self-compassion serves as internal framework for self-esteem.

Furthermore, when looking at difficulties in emotion regulation, a significant negative correlation was also found with self-esteem, meaning participants who had more trouble regulating emotions were more likely to have lower self-esteem; consistent with previous research that displays emotion dysregulation leads to an increase in negative self-appraisals, promoting feelings of helplessness and inadequacy (Crocker & Wolfe, 2001), and in-turn lowering self-esteem.

Overall, these findings confirm that self-compassion and difficulties in emotion regulation have an effect on self-esteem, supporting the present study's claim that these two factors are key internal contingencies of individual's self-esteem levels.

Internal Contingencies and ED Behaviour

Examining the three predictors of ED behaviours, the emergence of self-compassion as the only significant predictor directly contradicts previous research that would suggest self-esteem and difficulties in emotion regulation (Button, Sonuga-Barke, Davies & Thompson 1996; Gratz & Roemer, 2004) have strong predictive power in relation to ED behaviour.

This pattern indicates that the ability for individuals to respond to themselves with kindness and nonjudgement is a key factor in terms of reducing the prevalence of ED behaviours, such as restriction and binge-purging, supporting Neff's (2003) Self-Compassion Model. This model states that self-compassion is a proximal affect-regulation system, an internally generated process that soothes negative emotions and protects against harsh self-judgement. The pattern shown in the current hierarchical regression shows that self-esteem and difficulties in emotion regulation accounted for 12.8% of variance in ED behaviours, and within this the DERS emerged as the only significant predictor when the SCS was not accounted for, aligning with the view that low self-esteem and high difficulties in emotion regulation act as predisposing risk factors (Aldao, Nolen-Hoeksema, & Schweizer 2010).

When self-compassion was added into the model, explained variance rose to 26.7%, with SCS being a significant negative predictor. This suppression of significance strongly aligns with Neff's claim that self-compassion is the core mechanism regulating negative self-evaluations and intense emotions, suggesting while lower self-esteem predisposes individuals to interpret setbacks more internally, and difficulties in emotion regulation make them more likely to face more distress from these setbacks, these effects are downregulated by high self-compassion, reducing both self-critical thoughts (observed within individuals with low self-esteem) and intense emotions (strongly associated with the DERS; Jazaieri et al., 2013). Further evidence supports this interpretation, with laboratory studies confirming that lower self-compassion predicted an increase in dietary restraint and compensatory behaviours (e.g., purging), even after controlling for global self-esteem (Wasylikiw, MacKinnon, MacLellan, 2012), supporting the crucial role self-compassion plays in moderating ED behaviours.

These findings therefore suggest that clinical interventions for EDs may benefit from focusing on self-compassion targeted prevention and treatment measures, such as Compassion-Focused Therapy (*CFT*), as this may counteract both low self-esteem and

difficulties in emotion regulation simultaneously. Furthermore, for patients suffering from severe difficulties in emotion regulation it may be beneficial to incorporate self-compassion focused treatment approaches early-on, in attempt to prevent later development of ED behaviours.

Filling Theoretical Gaps

While the present study shows the important role of self-compassion in ED behaviours, future research should focus on two main areas in order to provide a more comprehensive understanding of self-esteem's role within this process.

Firstly, future work should expand on the role of contingencies of self-esteem by simultaneously measuring internal contingencies (such as self-acceptance) against external ones (such as appearance) to understand which ones are most predictive of ED behaviours. This will allow for more targeted clinical interventions by displaying whether focus should be placed on targeting internal or external contingencies.

Secondly, future work should specifically examine if improvements in self-compassion lead to subsequent increases in self-esteem, and in-turn reduce ED behaviours, possibly through a formal mediation analysis. This could be conducted through a longitudinal study in which participants are assigned to self-compassion practices (e.g., Mindful Self-Compassion; Neff, 2013), tracking changes in their SCS, RSES, and EDE-Q scores over time, in comparison to a control group. Establishing this mediation pathway would provide more clarification of the causal mechanisms between these factors, further informing clinical intervention focus.

Methodological limitations

One limitation with the present study is the utilisation of a cross-sectional design, meaning the results are correlational and cannot establish direct causations, and therefore potential confounds may account for the findings. Specifically, starvation has been shown to replicate behaviours associated with EDs, as well as lower self-esteem, self-compassion, and increase emotional dysregulation (Minnesota Starvation Study; Keys et al., 1950), displaying how other factors may influence the relationship between internal contingencies and ED behaviours. In future research it would therefore be beneficial to utilise an experimental design, directly manipulating self-esteem levels through variables such as criticism, and examining how these changes affect ED pathology so that a causal effect can be better established. Alternatively, utilising a longitudinal design to examine how RSES, DERS and SCS scores change over time in relation to EDE-Q scores would improve understanding of how these internal contingencies influence ED behaviours and provide more insight into the causal relationship.

It is also possible that the present study is capturing other confounding variables such as depression, which has implications in all measured variables within the present study, as high DERS scores may in fact reflect underlying depressive rumination, in turn lowering self-compassion and increasing severity of ED behaviours (Smith, Mason, Lavender, 2019). Therefore, future research should seek to also test overall mood and mental health, as means of ruling out these alternate explanations.

Moreover, the reliance on self-report questionnaires to measure the different constructs means participants may have inaccurately reported their answers as they were unaware of their true feelings and behaviours or wished to present more socially desirable responses. Future research could benefit from using ecological momentary assessments (EMA) to measure participants self-compassion, self-esteem, emotional, and ED thoughts

throughout the day, potentially reducing recall and desirability biases by capturing them in real-time and in a more natural context.

The reliance on volunteer sampling also carries the potential risks of self-selection bias, and additionally, the sample was disproportionately female and consisted solely of young adults, therefore limiting generalisability for several reasons. Firstly, female populations have been shown to have higher rates of lifetime ED prevalence (8.4% for women, 2.2% for men; Galmiche et al., 2019); secondly, onset prevalence is highest at age 21 (Ward et al., 2019); and finally, self-esteem has been shown to be lowest between the ages of 18-22 (Robins, Trzesniewski, Tracy, Gosling, 2002). Furthermore, the small male subsample limits the present study's ability to accurately detect gender effects. Future research should therefore aim to use a more representative sampling method (e.g., stratified sampling) across wider age ranges and non-student groups, in order to gather a more representative sample, therefore capturing if these internal contingencies of self-esteem operate similarly in a wider population.

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